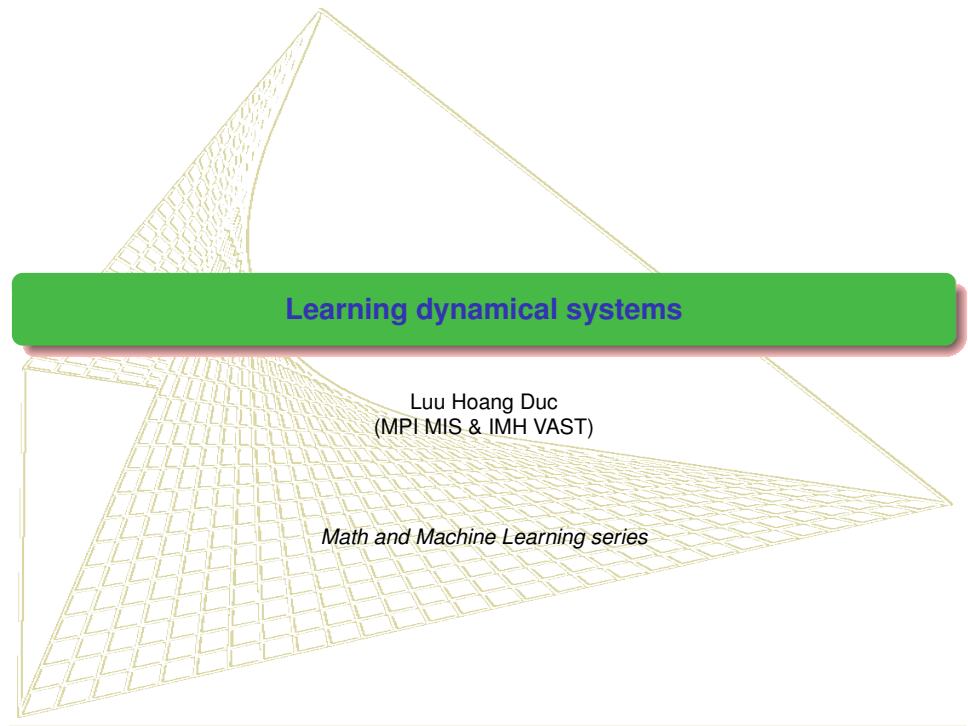




Learning dynamical systems

Luu Hoang Duc
(MPI MIS & IMH VAST)

Math and Machine Learning series

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Learning problem

- Given $\mathcal{D} = (X, Y) \in \mathbb{R}^m \times \mathbb{R}^d$, learn function $\hat{f} : X \rightarrow Y$ that minimize a loss function $L(X, Y)$.
- When f belongs to a class $\mathcal{H}$ the task can be written in the form

$$\hat{f} = \arg \min_{f \in \mathcal{H}} \left(\frac{1}{n} \sum_{i=1}^n \|f(x_i) - y_i\|^2 \right)$$

- When $\mathcal{H} = \{f_\theta : \theta \in \Theta\}$, then it is rewritten to the form

$$\hat{\theta} = \arg \min_{\theta \in \Theta} \left(\frac{1}{n} \sum_{i=1}^n \|f_\theta(x_i) - y_i\|^2 \right)$$

In practice, the target varies from trajectories to distributions.

- Baysian learning, statistical learning is a big topic...
- There is another data-driven approach using pathwise dynamics. This is the main content of this lecture.

Universal Approximation Theorems

Theorem 1 (Taylor, 1715). Any continuous function $f(x) : \mathbb{R} \rightarrow \mathbb{R}$ that is k -times differentiable at $a \in \mathbb{R}$ can be represented as a sum of polynomials:

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2}(x-a)^2 + \dots + \frac{f^{(k)}(a)}{k!}(x-a)^k + R_k(x) = \sum_{i=0}^k c_i(x-a)^i + R_k(x) \quad (13)$$

where $c_i = \frac{f^{(i)}(a)}{i!} = \frac{1}{i!} \frac{d^i}{dx^i} f(x)|_{x=a}$ and $R_k(x) = o(|x-a|^k)$ is the residual term.

Theorem 2 (Fourier, 1807). Any continuous and periodic function $f(x)$ can be expressed as a sum of sinusoids:

$$f(x) = A_0 + \sum_{i=1}^N A_i \cos\left(\frac{2\pi i}{T}x + \phi_i\right). \quad (15)$$

where T is the period of the function, A_i is the amplitude and ϕ_i is the phase of i^{th} harmonic component.

Theorem 3 (Weierstrass, 1885). Any continuous real-valued function $f(x) : [a, b] \rightarrow \mathbb{R}$ defined on the interval $[a, b]$ can be approximated with a polynomial function $p_N(x) = \sum_{i=0}^N c_i x^i$ with finite degree N such that:

$$|f(x) - p_N(x)| < \epsilon \quad (16)$$

for arbitrary $\epsilon > 0$.

variant of Hilbert's 13th problem [26]: Can any continuous function of more than two variables be expressed as a superposition of finitely many continuous functions of two variables? This problem was solved by Arnold and Kolmogorov in 1957 which resulted in the Kolmogorov-Arnold representation theorem [27] stated below:

Theorem 4 (Kolmogorov and Arnold, 1959). Any continuous multivariate function $f : [0, 1]^n \rightarrow \mathbb{R}$ can be written as

$$f(\mathbf{x}) = f(x_1, x_2, \dots, x_n) = \sum_{j=1}^{2n+1} \beta_j \left(\sum_{i=1}^n \alpha_{ij}(x_i) \right) \quad (17)$$

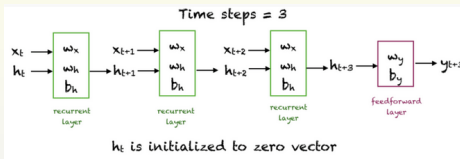
where $\alpha_{ij} : [0, 1] \rightarrow \mathbb{R}$ and $\beta_j : \mathbb{R} \rightarrow \mathbb{R}$.

Theorem 6 (Funahashi, Hornick et al., and Cybenko, 1989). Let $\mathbb{X}$ be any compact subset of $\mathbb{R}^n$ and σ be any sigmoid activation function, then the finite sum of the form:

$$f_{\text{NN}}(\mathbf{x}) = \mathbf{A}_2 \sigma(\mathbf{A}_1 \mathbf{x} + \mathbf{b}_1) = \sum_{j=1}^{m_1} a_{2j} \sigma(\mathbf{A}_{1j} \mathbf{x} + b_{1j}) \quad (20)$$

is dense in $\mathbb{X}$. In other words, given any $f : \mathbb{X} \rightarrow \mathbb{R}$ and $\epsilon > 0$, there is a finite sum: f_{NN} as in Eq. (20) for which $|f(\mathbf{x}) - f_{\text{NN}}(\mathbf{x})| < \epsilon$ for all $\mathbf{x} \in \mathbb{X}$.

Recurrent Neural networks



$$y_T = f(W_y h_T + b_y); \quad h_{t+1} = \sigma(W_h h_t + W_x x_t + b_h). \quad (1.1)$$

Theorem 5 (Universal approximation theorem of deep RNN 1) Let $f : \mathbb{R}^{d_x \times N} \rightarrow \mathbb{R}^{d_y \times N}$ be a continuous past-dependent sequence-to-sequence function and σ be a non-degenerate activation function. Then, for any $\epsilon > 0$ and compact subset $K \subset \mathbb{R}^{d_x}$, there exists a deep RNN $\mathcal{N}$ of width $d_x + d_y + 3 + \alpha(\sigma)$ such that

$$\sup_{x \in K^N} \sup_{1 \leq t \leq N} \|f(x)[t] - \mathcal{N}(x)[t]\| < \epsilon, \quad (16)$$

where $\alpha(\sigma) = \begin{cases} 0 & \sigma \text{ is ReLU or a non-degenerating function with } \sigma(z_0) = 0. \\ 1 & \sigma \text{ is a non-degenerating function with } \sigma(z_0) \neq 0. \end{cases}$

Network	Function class	Activation	Result
RNN	$C(\mathcal{K}, \mathbb{R}^{d_y})^\dagger$	ReLU	$w_{\min} \leq d_x + d_y + 3$
		conti. nonpoly ¹	$w_{\min} \leq d_x + d_y + 3$
		conti. nonpoly ²	$w_{\min} \leq d_x + d_y + 4$
	$L^p(\mathcal{K}, \mathbb{R}^{d_y})^\dagger$	conti. nonpoly ^{1,2}	$w_{\min} \leq \max\{d_x + 1, d_y\} + 1$
	$L^p(\mathbb{R}^{d_x}, \mathbb{R}^{d_y})^\dagger$	ReLU	$w_{\min} = \max\{d_x + 1, d_y\}$
LSTM	$C(\mathcal{K}, \mathbb{R}^{d_y})^\dagger$		$w_{\min} \leq d_x + d_y + 3$
	$L^p(\mathcal{K}, \mathbb{R}^{d_y})^\dagger$		$w_{\min} \leq \max\{d_x + 1, d_y\} + 1$
GRU	$C(\mathcal{K}, \mathbb{R}^{d_y})^\dagger$		$w_{\min} \leq d_x + d_y + 3$
	$L^p(\mathcal{K}, \mathbb{R}^{d_y})^\dagger$		$w_{\min} \leq \max\{d_x + 1, d_y\} + 1$

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ORIGINAL CONTRIBUTION

Approximation of Dynamical Systems by Continuous Time Recurrent Neural Networks

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The dynamics of the continuous time recurrent neural network with m units, which is discussed in this paper, is described by the following system of ordinary differential equations:

$$\frac{du_i(t)}{dt} = -\frac{u_i(t)}{\tau_i} + \sum_{j=1}^m w_{ij} \sigma(u_j(t)) + I_i(t) \quad (i = 1, \dots, m), \quad (1)$$

where $u_i(t)$ is the internal state of the i -th unit, τ_i is the time constant of the i -th unit, w_{ij} are connection weights, $I_i(t)$ is the input to the i -th unit, and $\sigma(u_i(t))$ is the output of the i -th unit. Here, σ is called the output function, and C^1 -sigmoid functions (nonconstant, bounded, and monotone increasing C^1 -functions) are used. As σ ,

$$\sigma(x) = 1/(1 + \exp(-x)) \quad (2)$$

is usually used.

In the following, we deal with recurrent neural networks with the same time constant τ (i.e., $\tau_i = \tau$) and without input (i.e., $I_i(t) = 0$). We set $u(t) = (u_1(t), \dots, u_m(t))$ and use $W = (w_{ij})$ as an $m \times m$ weight matrix. Let $\sigma: \mathbb{R}^m \rightarrow \mathbb{R}^m$ be denoted by a sigmoid mapping

$$\sigma^{(t)}(u_1, \dots, u_m) = (\sigma(u_1), \dots, \sigma(u_m)) \quad (3)$$

then the vector expression of eqn (1) is

$$\dot{u}(t) = -\frac{1}{\tau} u(t) + W \sigma(u(t)). \quad (4)$$

3. APPROXIMATE REALIZATION THEOREMS OF DYNAMICAL SYSTEM TRAJECTORIES

Let the point of the n -dimensional Euclidean space $\mathbb{R}^n$ be denoted by $x = (x_1, \dots, x_n)$ and the Euclidean norm of x be defined by $|x|$.

The dynamical system on an open set of $\mathbb{R}^n$ means a system defined by an autonomous ordinary differ-

ential equation $\dot{x}(t) = f(x(t))$ and an appropriate initial state $x(0)$ of the system.

$$\max_{t \in I} |x(t) - u(t)| < \epsilon \quad (6)$$

holds, where $u(t) = (u_1(t), \dots, u_m(t))$ is the internal state of output units of the network.

For dynamical systems, the condition for $\hat{K}$ is satisfied a priori by definition, so we can restate Theorem 1 as follows.

THEOREM 2. Let $D \subset \mathbb{R}^n$ and $F: D \rightarrow \mathbb{R}^n$ be the same as above, and suppose that $\hat{x} = F(x)$ defines a dynamical system on D . Let K be a compact subset of D and we consider trajectories of the system on the interval $I = [0, T]$ ($0 < T < \infty$). Then, for an arbitrary $\epsilon > 0$, there exist an integer N and a recurrent neural network with n output units and N hidden units such that for any trajectory $\{x(t); 0 \leq t \leq T\}$ of the system with initial value $x(0) \in K$ and an appropriate initial state of the network,

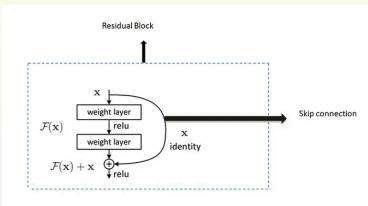
$$\max_{t \in I} |x(t) - u(t)| < \epsilon$$

holds, where $u(t) = (u_1(t), \dots, u_m(t))$ is the internal state of output units of the network.

As a corollary of Theorem 2, we obtain the following:

COROLLARY 1. Let σ be a strictly increasing C^1 -sigmoid function such that $\sigma(\mathbb{R}) = (0, 1)$. Let D be an open subset of $(0, 1)^n$, $F: D \rightarrow \mathbb{R}^n$ be a C^1 -mapping, and suppose that $\hat{x} = F(x)$ defines a dynamical system on D . Let K be a compact subset of D and we consider trajectories of the system on interval $I = [0, T]$ ($0 < T < \infty$). Then, for an arbitrary $\epsilon > 0$, there exist an integer N and a recurrent neural network with n output units and N hidden units such that for any trajectory $\{x(t); 0 \leq t \leq T\}$ of the system with initial value $x(0) \in K$ and an appropriate initial state of the network,

Residual Neural Networks



- ReSNet update

$$y_{i+1} = y_i + \sigma(\theta_i y_i). \quad (1.2)$$

- Chen et al. (2018): ResNet with infinite layers = neural ODE!!! View (1.2) as a step of the Euler scheme for a controlled ODE

$$\dot{y}_t = \sigma(\theta_t y_t), \quad y_0 \in \mathbb{R}^{d_0}. \quad (1.3)$$

- Kidger - Chen - Lyons (2021): view it as a controlled differential equation

$$h_{t+1} = h_t + f_\theta(h_t + b_t) dX_t \Rightarrow dh_t = f_\theta(h_t + b_t) dX_t \quad (1.4)$$

which is interpreted as a rough differential/difference equation.

- Bayer-Friz-Tapia (2023): Stability of ResNet

$$\sup_{k=0..n} \|h_k - \tilde{h}_k\| \leq C'_{p,n} (\|x_0 - \tilde{x}_0\| + \rho_p(\mathbf{x}, \tilde{\mathbf{x}})) \quad (1.5)$$

Gradient decent method

- Minimization problem

$$\min_{\theta \in \mathbb{R}^d} Q(\theta), \quad Q \in C^1. \quad (1.6)$$

- Global solution: $\theta^* \in \arg \min_{\theta \in \mathbb{R}^d} Q(\theta) \Rightarrow \nabla Q(\theta^*) = 0$. This leads to an approximation scheme to achieve θ^* , which is the zero solution of $\nabla Q(\theta) = 0$

$$\theta_{t+1} = \theta_t - \eta \nabla Q(\theta_t), \quad (1.7)$$

for the learning rate $\eta > 0$. (1.7) is called Gradient descent method.

- When η small enough, (1.7) can also be viewed as the Euler scheme for solving the ODE

$$\dot{\theta} = -\nabla Q(\theta), \quad (1.8)$$

where one attempts to solve for the steady state θ^* of (1.8).

Stochastic gradient descent

- Minimization problem

$$\min_{\theta \in \mathbb{R}^d} Q(\theta) = \min_{\theta \in \mathbb{R}^d} \frac{1}{m} \sum_{i=1}^m Q^{(i)}(\theta), \quad Q^{(i)} \in \mathcal{C}^1. \quad (1.9)$$

When m large enough, (1.7) is replaced by: at each step t choose a random set Ω_t of $\{1, \dots, m\}$ (with or without replacement) and then updating

$$\theta_{t+1} := \theta_t - \eta \frac{1}{|\Omega_t|} \sum_{i \in \Omega_t} \nabla Q^{(i)}(\theta_t) = \theta_t - \eta \nabla \tilde{Q}_t(\theta_t) \quad (1.10)$$

No matter how random the choice, can rewrite SGD recursion

$$\theta_{t+1} = \theta_t - \eta \nabla Q(\theta_t) + \eta^\alpha \epsilon_t(\theta_t). \quad (1.11)$$

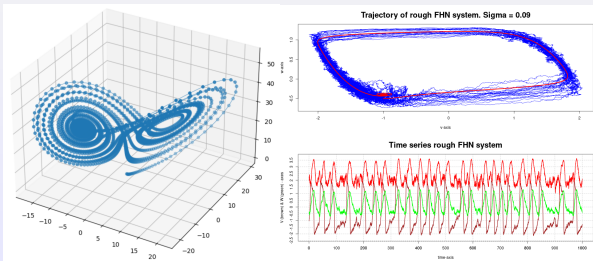
- (1.11) can also be viewed as the Euler scheme of the "limiting equation"

$$d\theta = -\nabla Q(\theta)dt + \sigma(\theta)dX_t, \quad \text{where } \sigma\sigma^T = \eta^{2\alpha} \mathbb{E}[\epsilon\epsilon^T | \theta_t = \theta] \quad (1.12)$$

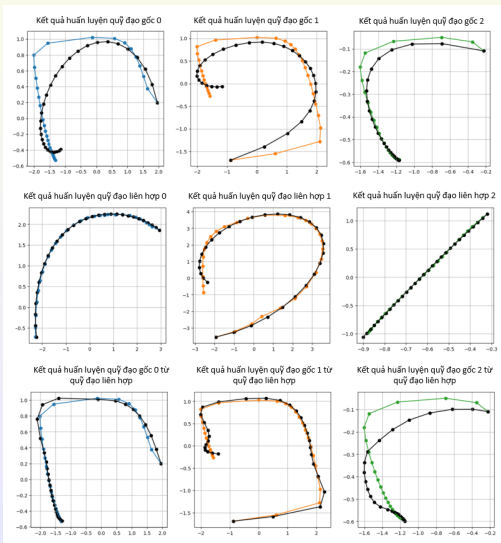
- If Ω_t i.i.d. then CLT gives $\alpha = \frac{1}{2}$ and X -Gaussian. But not necessary i.i.d.
- Questions: How does the distribution of (1.11) converge to some limit? And which calculus is used to interpret and solve the limiting equation?
- Work by Hairer et al. on ergodicity if X is fractional Brownian motion. But we prefer an approach which is purely discrete and data-driven.

Data driven dynamical systems

- A (continuous) dynamical system on a metric space D is the triple $(\mathbb{R}, D, \pi)$ where $\pi : \mathbb{R} \times X \rightarrow X$, $\pi(0, x_0) = x_0$, $\pi(t + s, x_0) = \pi(t, \pi(s, x_0))$, $\forall t, s \in \mathbb{R}, x_0 \in D$.
- Abstract form: $\frac{d}{dt}y_t = f_\theta(t, y_t)$ or $y_{k+1} = F_\theta(y_k)$ where $\theta \in \Theta$ is a (multi-dimensional) parameter. When $f_\theta(t, y_t; \theta) = f_\theta(y_t)$ it is autonomous
- Examples: Lorenz system, Fitzhugh Nagumo neuro-system, logistic map.
- Stability theory is important! For Feedforward NN it does not play a role. But RNN, the recurrent unit is better stable.
- Challenges: unknown dynamics (recover governing equations), future state prediction, non-linearity, high dimensionality (need dimension reduction with appropriate coordinates), stochastically perturbed by noises, asymptotic stability (able to approximate in short term but failed to predict long term due to error accumulation).



Fitzhugh Nagumo-Conjugacy method



16 Dec 2020

Denoising Diffusion Probabilistic Models

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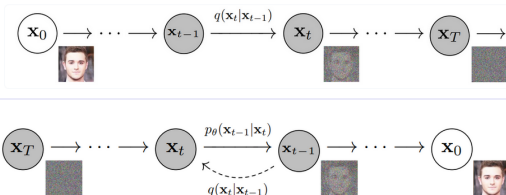
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Abstract

We present high quality image synthesis results using diffusion probabilistic models, a class of latent variable models inspired by considerations from nonequilibrium thermodynamics. Our best results are obtained by training on a weighted variational bound designed according to a novel connection between diffusion probabilistic models and denoising score matching with Langevin dynamics, and our models naturally admit a progressive lossy decompression scheme that can be interpreted as a generalization of autoregressive decoding. On the unconditional CIFAR10 dataset,



Diffusion model

$$q(\mathbf{x}_{1:T}|\mathbf{x}_0) := \prod_{t=1}^T q(\mathbf{x}_t|\mathbf{x}_{t-1}) := \prod_{t=1}^T \mathcal{N}(\mathbf{x}_t; \sqrt{1 - \beta_t}\mathbf{x}_{t-1}, \beta_t \mathbf{I})$$

Where $\beta_1, \dots, \beta_T$ is a variance schedule (either learned or fixed) which, if well-behaved, **ensures that x_T is nearly an isotropic Gaussian for sufficiently large T .**

$$\mathcal{X}_t \sim \mathcal{N}(\mathcal{X}_{t-1}, 1) \iff \mathcal{X}_t = \mathcal{X}_{t-1} + N(0, 1)$$

$$p_{\mathcal{X}_t}(x_t) = \int p(x_t|x_{t-1})p(x_{t-1})dx_{t-1}$$

$$p_{\theta}(\mathbf{x}_{0:T}) := p(\mathbf{x}_T) \prod_{t=1}^T p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t) := p(\mathbf{x}_T) \prod_{t=1}^T \mathcal{N}(\mathbf{x}_{t-1}; \boldsymbol{\mu}_{\theta}(\mathbf{x}_t, t), \boldsymbol{\Sigma}_{\theta}(\mathbf{x}_t, t))$$

$$p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t) := \mathcal{N}(\mathbf{x}_{t-1}; \boldsymbol{\mu}_{\theta}(\mathbf{x}_t, t), \boldsymbol{\Sigma}_{\theta}(\mathbf{x}_t, t))$$

$$\mathbb{E}[-\log p_{\theta}(\mathbf{x}_0)] \leq \mathbb{E}_q \left[-\log \frac{p_{\theta}(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \right] =: L_{vlb}$$

$$L_{vlb} = L_0 + L_1 + \dots + L_{T-1} + L_T$$

$$L_0 = -\log p_{\theta}(x_0|x_1)$$

$$L_{t-1} = D_{KL}(q(x_{t-1}|x_t, x_0) || p_{\theta}(x_{t-1}|x_t))$$

$$L_T = D_{KL}(q(x_T|x_0) || p(x_T))$$

Diffusion model

A nice property of the above process is that we can sample $\mathbf{x}_t$ at any arbitrary time step t in a closed form using reparameterization trick. Let $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{i=1}^t \alpha_i$:

It is noteworthy that the reverse conditional probability is tractable when conditioned on $\mathbf{x}_0$:

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_{t-1}; \tilde{\boldsymbol{\mu}}(\mathbf{x}_t, \mathbf{x}_0), \tilde{\boldsymbol{\beta}}_t \mathbf{I})$$

$$\tilde{\beta}_t = 1 / \left(\frac{\alpha_t}{\beta_t} + \frac{1}{1 - \bar{\alpha}_{t-1}} \right) = 1 / \left(\frac{\alpha_t - \bar{\alpha}_t + \beta_t}{\beta_t(1 - \bar{\alpha}_{t-1})} \right) = \frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t} \cdot \beta_t$$

$$\begin{aligned} \tilde{\boldsymbol{\mu}}_t &= \frac{\sqrt{\bar{\alpha}_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} \mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1 - \bar{\alpha}_t} \frac{1}{\sqrt{\bar{\alpha}_t}} (\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t} \boldsymbol{\epsilon}_t) \\ &= \frac{1}{\sqrt{\bar{\alpha}_t}} \left(\mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \boldsymbol{\epsilon}_t \right) \end{aligned}$$

$$\begin{aligned} L_t^{\text{simple}} &= \mathbb{E}_{t \sim [1, T], \mathbf{x}_0, \boldsymbol{\epsilon}_t} \left[\|\boldsymbol{\epsilon}_t - \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)\|^2 \right] \\ &= \mathbb{E}_{t \sim [1, T], \mathbf{x}_0, \boldsymbol{\epsilon}_t} \left[\|\boldsymbol{\epsilon}_t - \boldsymbol{\epsilon}_\theta(\sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \boldsymbol{\epsilon}_t, t)\|^2 \right] \end{aligned}$$

Neural Ordinary Differential Equations

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Abstract

We introduce a new family of deep neural network models. Instead of specifying a discrete sequence of hidden layers, we parameterize the derivative of the hidden state using a neural network. The output of the network is computed using a black-box differential equation solver. These continuous-depth models have constant memory cost, adapt their evaluation strategy to each input, and can explicitly trade numerical precision for speed. We demonstrate these properties in continuous-depth residual networks and continuous-time latent variable models. We also construct continuous normalizing flows, a generative model that can train by maximum likelihood, without partitioning or ordering the data dimensions. For training, we show how to scalably backpropagate through any ODE solver, without access to its internal operations. This allows end-to-end training of ODEs within larger models.

1 Introduction

Models such as residual networks, recurrent neural network decoders, and normalizing flows build complicated transformations by composing a sequence of transformations to a hidden state:

$$\mathbf{h}_{t+1} = \mathbf{h}_t + f(\mathbf{h}_t, \theta_t) \quad (1)$$

where $t \in \{0, \dots, T\}$ and $\mathbf{h}_t \in \mathbb{R}^D$. These iterative updates can be seen as an Euler discretization of a continuous transformation (Lu et al., 2017; Haber and Ruthotto, 2017; Ruthotto and Haber, 2018).

What happens as we add more layers and take smaller steps? In the limit, we parameterize the continuous dynamics of hidden units using an ordinary differential equation (ODE) specified by a neural network:

$$\frac{d\mathbf{h}(t)}{dt} = f(\mathbf{h}(t), t, \theta) \quad (2)$$

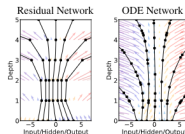


Figure 1: *Left*: A Residual network defines a discrete sequence of finite transformations. *Right*: A ODE network defines a vector field, which continuously transforms the state. *Both*: Circles represent evaluation locations.

$$\frac{d\mathbf{h}(t)}{dt} = f_{\theta}(\mathbf{h}(t), t) \quad (2.1)$$

where $\mathbf{h}(t) \in \mathbb{R}^D$ state at t , $f_{\theta}(\cdot, \cdot)$ is a neural network θ . In NeuralODE, hidden layer dimension is constant. Input NeuralODE is the initial state $\mathbf{h}(0)$ and output NeuralODE is the state $\mathbf{h}(T)$. NeuralODE solves the IVP with vectorfield f_{θ} at time $[0, T]$.

1.1.2 What is a neural differential equation anyway?

A *neural differential equation* is a differential equation using a neural network to parameterise the vector field. The canonical example is a *neural ordinary differential equation* [Che+18b]:

$$y(0) = y_0 \quad \frac{dy}{dt}(t) = f_{\theta}(t, y(t)).$$

Here θ represents some vector of learnt parameters, $f_{\theta}: \mathbb{R} \times \mathbb{R}^{d_1 \times \dots \times d_k} \rightarrow \mathbb{R}^{d_1 \times \dots \times d_k}$ is any standard neural architecture, and $y: [0, T] \rightarrow \mathbb{R}^{d_1 \times \dots \times d_k}$ is the solution. For many applications f_{θ} will just be a simple feedforward network.

The central idea now is to use a differential equation solver as part of a learnt differentiable computation graph (the sort of computation graph ubiquitous to deep learning).

Adjoint sensitivity method for NeuralODE. Consider loss function $L()$ with input to be the output of Neural ODE

$$L(\mathbf{z}(t_1)) = L\left(\mathbf{z}(t_0) + \int_{t_0}^{t_1} f_{\theta}(\mathbf{z}(t), t) dt\right) = L(\text{ODESolve}(\mathbf{z}(t_0), f_{\theta}, t_0, t_1)) \quad (2.2)$$

Neural ODE-Loss function

Consider optimizing a scalar-valued loss function $L(\cdot)$, whose input is the result of an ODE solver:

$$L(\mathbf{z}(t_1)) = L\left(\mathbf{z}(t_0) + \int_{t_0}^{t_1} f(\mathbf{z}(t), t, \theta) dt\right) = L(\text{ODESolve}(\mathbf{z}(t_0), f, t_0, t_1, \theta)) \quad (3)$$

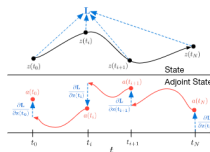


Figure 2: Reverse-mode differentiation of an ODE solution. The adjoint sensitivity method solves an augmented ODE backwards in time. The augmented system contains both the original state and the sensitivity of the loss with respect to the state. If the loss depends directly on the state at multiple observation times, the adjoint state must be updated in the direction of the partial derivative of the loss with respect to each observation.

To optimize L , we require gradients with respect to θ . The first step is to determine how the gradient of the loss depends on the hidden state $\mathbf{z}(t)$ at each instant. This quantity is called the *adjoint* $\mathbf{a}(t) = \partial L / \partial \mathbf{z}(t)$. Its dynamics are given by another ODE, which can be thought of as the instantaneous analog of the chain rule:

$$\frac{d\mathbf{a}(t)}{dt} = -\mathbf{a}(t)^\top \frac{\partial f(\mathbf{z}(t), t, \theta)}{\partial \mathbf{z}} \quad (4)$$

We can compute $\partial L / \partial \mathbf{z}(t_0)$ by another call to an ODE solver. This solver must run backwards, starting from the initial value of $\partial L / \partial \mathbf{z}(t_1)$. One complication is that solving this ODE requires the knowing value of $\mathbf{z}(t)$ along its entire trajectory. However, we can simply recompute $\mathbf{z}(t)$ backwards in time together with the adjoint, starting from its final value $\mathbf{z}(t_1)$.

Computing the gradients with respect to the parameters θ requires evaluating a third integral, which depends on both $\mathbf{z}(t)$ and $\mathbf{a}(t)$:

$$\frac{dL}{d\theta} = - \int_{t_1}^{t_0} \mathbf{a}(t)^\top \frac{\partial f(\mathbf{z}(t), t, \theta)}{\partial \theta} dt \quad (5)$$

The vector-Jacobian products $\mathbf{a}(t)^\top \frac{\partial f}{\partial \theta}$ and $\mathbf{a}(t)^\top \frac{\partial f}{\partial \mathbf{z}}$ in (4) and (5) can be efficiently evaluated by automatic differentiation, at a time cost similar to that of evaluating f . All integrals for solving $\mathbf{z}$, $\mathbf{a}$

Input: dynamics parameters θ , start time t_0 , stop time t_1 , final state $\mathbf{z}(t_1)$, loss gradient $\partial L / \partial \mathbf{z}(t_1)$

$$s_0 = [\mathbf{z}(t_1), \frac{\partial L}{\partial \mathbf{z}(t_1)}, \mathbf{0}_{|\theta|}]$$

def aug_dynamics $[\mathbf{z}(t), \mathbf{a}(t), \cdot], t, \theta$:

return $[f(\mathbf{z}(t), t, \theta), -\mathbf{a}(t)^\top \frac{\partial f}{\partial \mathbf{z}}, -\mathbf{a}(t)^\top \frac{\partial f}{\partial \theta}]$

$$[\mathbf{z}(t_0), \frac{\partial L}{\partial \mathbf{z}(t_0)}, \frac{\partial L}{\partial \theta}] = \text{ODESolve}(s_0, \text{aug_dynamics}, t_1, t_0, \theta)$$

return $\frac{\partial L}{\partial \mathbf{z}(t_0)}, \frac{\partial L}{\partial \theta}$

▷ Define initial augmented state

▷ Define dynamics on augmented state

▷ Compute vector-Jacobian products

▷ Solve reverse-time ODE

▷ Return gradients

Neural differential equations

Neural ODE-RNN encoder and ODEsolve

a global set of latent dynamics shared across all time series. Given observation times $t_0, t_1, \dots, t_N$ and an initial state $\mathbf{z}_{t_0}$, an ODE solver produces $\mathbf{z}_{t_1}, \dots, \mathbf{z}_{t_N}$, which describe the latent state at each observation. We define this generative model formally through a sampling procedure:

$$\mathbf{z}_{t_0} \sim p(\mathbf{z}_{t_0}) \quad (11)$$

$$\mathbf{z}_{t_1}, \mathbf{z}_{t_2}, \dots, \mathbf{z}_{t_N} = \text{ODEsolve}(\mathbf{z}_{t_0}, f, \theta_f, t_0, \dots, t_N) \quad (12)$$

$$\text{each } \mathbf{x}_{t_i} \sim p(\mathbf{x}|\mathbf{z}_{t_i}, \theta_x) \quad (13)$$

Function f is a time-invariant function that takes the value $\mathbf{z}$ at the current time step and outputs the gradient: $\partial \mathbf{z}(t)/\partial t = f(\mathbf{z}(t), \theta_f)$. We parametrize this function using a neural net. Because f is time-

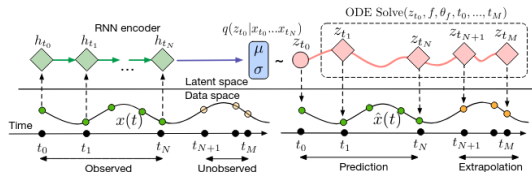
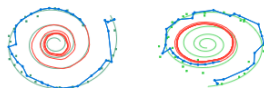
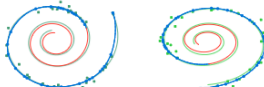


Figure 6: Computation graph of the latent ODE model.



(a) Recurrent Neural Network



(b) Latent Neural Ordinary Differential Equation

— Ground Truth
● Observation
— Prediction
— Extrapolation



(c) Latent Trajectories

“Hey, that’s not an ODE”: Faster ODE Adjoints via Seminorms

Patrick Kidger^{1,2} Ricky T. Q. Chen^{3,4} Terry Lyons^{1,2}

Abstract

Neural differential equations may be trained by backpropagating gradients via the adjoint method, which is another differential equation typically solved using an adaptive-step-size numerical differential equation solver. A proposed step is accepted if its error, *relative to some norm*, is sufficiently small; else it is rejected, the step is shrunk, and the process is repeated. Here, we demonstrate that the particular structure of the adjoint equations makes the usual choices of norm (such as L^2) unnecessarily stringent. By replacing it with a more appropriate (semi)norm, fewer steps are unnecessarily rejected and the backpropagation is made faster. This requires only minor code modifications. Experiments on a wide range of tasks—including time series, generative modeling, and physical control—demonstrate a median improvement of 40% fewer function evaluations. On some problems we see as much as 62% fewer function evaluations, so that the overall training time is roughly halved.

1. Introduction

We begin by recalling the usual set-up for neural differential equations.

are composed such that

$$\begin{aligned} z(\tau) &= \ell_1(x, \phi), \\ z(t) &= z(\tau) + \int_{\tau}^t f(s, z(s), \theta) \, ds \\ y &\approx \ell_2(z(T), \psi). \end{aligned} \tag{1}$$

The variables ϕ, θ, ψ denote learnable parameters and the ODE is solved over the interval $[\tau, T]$. We include the (often linear) maps $\ell_1(\cdot, \phi), \ell_2(\cdot, \psi)$ for generality, as in many contexts they are important for the expressiveness of the model (Dupont et al., 2019; Zhang et al., 2020), though our contributions will be focused around the ODE component and will not depend on these maps.

Here we will consider neural differential equations that may be interpreted as a neural ODE.

1.2. Applications

Neural differential equations have, to the best of our knowledge, three main applications:

1. Time series modeling. Rubanova et al. (2019) interleave Neural ODEs with RNNs to produce ODEs with jumps. Kidger et al. (2020) take $f(t, z, \theta) = g(z, \theta) \frac{dX}{dt}(t)$, dependent on some time-varying input X , to produce a neural *controlled* differential equation.

Neural Controlled Differential Equations for Irregular Time Series

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Abstract

Neural ordinary differential equations are an attractive option for modelling temporal dynamics. However, a fundamental issue is that the solution to an ordinary differential equation is determined by its initial condition, and there is no mechanism for adjusting the trajectory based on subsequent observations. Here, we demonstrate how this may be resolved through the well-understood mathematics of controlled differential equations. The resulting *neural controlled differential equation model* is directly applicable to the general setting of partially-observed irregularly-sampled multivariate time series, and (unlike previous work on this problem) it may utilise memory-efficient adjoint-based backpropagation even across observations. We demonstrate that our model achieves state-of-the-art performance against similar (ODE or RNN based) models in empirical studies on a range of datasets. Finally we provide theoretical results demonstrating universal approximation, and that our model subsumes alternative ODE models.

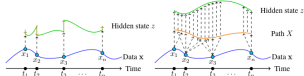


Figure 1: Some data process is observed at times $t_1, \dots, t_n$ to give observations $x_1, \dots, x_n$. It is otherwise unobserved. **Left:** Previous work has typically modified hidden state at each observation, and perhaps continuously evolved the hidden state between observations. **Right:** In contrast, the hidden state of the Neural CDE model has continuous dependence on the observed data.

3 Method

Suppose for simplicity that we have a fully-observed but potentially irregularly sampled time series $\mathbf{x} = \{(t_0, x_0), (t_1, x_1), \dots, (t_n, x_n)\}$, with each $t_i \in \mathbb{R}$ the timestamp of the observation $x_i \in \mathbb{R}^d$, and $t_0 < \dots < t_n$. (We will consider partially-observed data later.)

Let $X: [t_0, t_n] \rightarrow \mathbb{R}^{d+1}$ be the natural cubic spline with knots at $t_0, \dots, t_n$ such that $X_{t_i} = (x_i, t_i)$. As $\mathbf{x}$ is often assumed to be a discretisation of an underlying process, observed only through $\mathbf{x}$, then X is an approximation to this underlying process. Natural cubic splines have essentially the minimum regularity for handling certain edge cases; see Appendix A for the technical details.

Let $f_\theta: \mathbb{R}^d \rightarrow \mathbb{R}^{d \times (d+1)}$ be any neural network model depending on parameters θ . The value w is a hyperparameter describing the size of the hidden state. Let $G_\theta: \mathbb{R}^{d+1} \rightarrow \mathbb{R}^w$ be any neural network model depending on parameters θ .

Then we define the *neural controlled differential equation model* as the solution of the CDE

$$z_t = z_{t_0} + \int_{t_0}^t f_\theta(z_s) dX_s \quad \text{for } t \in [t_0, t_n], \quad (3)$$

where $z_t = (z_t^1, \dots, z_t^w) \in \mathbb{R}^w$. This initial condition is used to avoid translational invariance. Analogous

Motivation from stochastic calculus

- path-wise RDEs on $(\Omega, \mathcal{F}, \mathbb{P}, (\theta_t)_{t \in R})$: Sell, Oseledets, Millionschikov, Kloeden, Arnold...

$$\dot{y} = f(\theta_t \omega, y). \quad (2.3)$$

- SDEs with Brownian motions: Itô & Stratonovich calculus. Arnold, Kloeden, Caraballo, Schmalfuss, Crauel, Scheutzow...

$$dy_t = f(y_t)dt + g(y_t)dW_t. \quad (2.4)$$

- L.C. Young (1936). An integration of Hölder type, connected with Stieltjes integration. Acta Math.
- K. T. Chen (1954). "Iterated Integrals and Exponential Homomorphisms". Proceedings of the London Mathematical Society.
- H. J. Sussmann (1978). On the gap between deterministic and stochastic ordinary differential equations. Annals of Prob.
- T. Lyons (1998). Differential equations driven by rough signals. Revista Matematica Iberoamericana.
- SDEs with fractional Brownian motions: rough differential equations. Rough path theory (Lyons 1998), Gubinelli, Friz, Nualart, Hairer...

$$dy_t = f(y_t)dt + g(y_t)dB_t^H, H \in (\frac{1}{4}, 1). \quad (2.5)$$

- Solution Itô-Lyons map: $IL(\omega) = y(T, \omega, y_0)$ of random ODEs and SDEs is measurable w.r.t. ω . But $IL(\omega)$ is continuous w.r.t. ω (Lyons' universal theorem).

Signatures of rough paths

- Let $x \in C^{1-var}(I, \mathbb{R}^d)$. x_t^i is the coordinate i of x_t at time t . Define

$$\mathbf{g}^{k;i_1, \dots, i_k} := \int_s^t \int_s^{u_k} \dots \int_s^{u_2} dx_{u_1}^{i_1} \dots dx_{u_k}^{i_k}.$$

and the set

$$\mathbf{g} = \left(\mathbf{g}^{k;i_1, \dots, i_k} : 1 \leq k \leq N; i_1, \dots, i_k \in \{1, \dots, d\} \right)$$

to be the N -level signature of $x|_{[s,t]}$ and is written as $S_N(x)_{s,t}$. In other words,

$$\begin{aligned} S_N(x)_{s,t} &\equiv \left(1, \int_{s < u < t} dx_u, \dots, \int_{s < u_1 < \dots < u_k < t} dx_{u_1} \otimes \dots \otimes dx_{u_k} \right) \\ &\in \oplus_{k=0}^N (\mathbb{R}^d)^{\otimes k} =: \mathcal{T}_N(\mathbb{R}^d), \end{aligned}$$

- Then $S_N(x)_{s, \cdot}$ satisfies the controlled DE by x as follows

$$\begin{cases} dS_N(x)_{s,t} = S_N(x)_{s,t} \otimes dx_t, \\ S_N(x)_{s,s} = 1. \end{cases}$$

- Rough differential equation

$$dy = V(y)dx \Rightarrow y_{s,t} \approx \tilde{V}(y_s) \otimes S_N(x)_{s,t}$$

- fractional Brownian motion (fBm) with Hurst parameter $H \in (0, 1)$: a continuous centered Gaussian process $B^H = (B^H(t), t \in \mathbb{R})$, with covariance function

$$R(s, t) = \frac{1}{2}(|t|^{2H} + |s|^{2H} - |t - s|^{2H}), \quad s, t \in \mathbb{R},$$

When $H = 1/2$, B is the standard Brownian motion.

- B^H is neither Markov nor semi-martingale if $H \neq \frac{1}{2}$!
- Kolmogorov-Centsov theorem: B_t^H is H^- - Hölder continuity a.s. in t .
- For simplicity, think of $dy = g(y)dx$ with scalar path x . First solve the ODE $\dot{u} = g(u)$ to obtain solution $\pi(t, y_0)$ and then assign $y_t = \pi(x_t, y_0)$. Then

$$dy_t = \frac{\partial \pi}{\partial x}(x_t, y_0) dx_t = g(\pi(x_t, y_0)) dx_t = g(y_t) dx_t.$$

By Taylor expansion

$$\begin{aligned} y_{a,b} &= \frac{\partial \pi}{\partial x}(x_a, y_0) x_{a,b} + \frac{1}{2} \frac{\partial^2 \pi}{\partial x^2}(x_a, y_0) (x_{a,b})^2 + \dots \\ &= g(\pi(x_a, y_0)) x_{a,b} + \frac{1}{2} \frac{\partial}{\partial x} g(\pi(x_a, y_0)) (x_{a,b})^2 + \dots \\ &= g(y_a) x_{a,b} + \frac{1}{2} Dg(y_a) g(y_a) (x_{a,b})^2 + \dots \end{aligned}$$

One needs more information from higher order terms if x is of low regularity.

Rough paths

In the simplest form for $\beta \in (\frac{1}{3}, \frac{1}{2})$, a couple $\mathbf{x} = (x, \mathbb{X})$, with $x \in C^\beta([a, b], \mathbb{R}^d)$ and $\mathbb{X} \in C_2^{2\beta}([a, b]^2, \mathbb{R}^d \otimes \mathbb{R}^d)$ is called a *rough path* if they satisfy Chen's relation

$$\mathbb{X}_{s,t} - \mathbb{X}_{s,u} - \mathbb{X}_{u,t} = x_{s,u} \otimes x_{u,t}, \quad \forall a \leq s \leq u \leq t \leq b. \quad (2.6)$$

We write $\mathbb{X}_{s,t} = (x \otimes x)_{s,t} =: \int_s^t x_{s,u} \otimes dx_u$ and call it Lévy area of x .

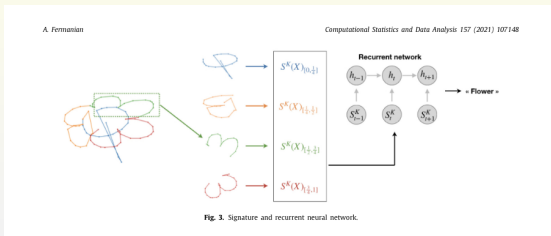
- $X_t = \int_0^t a_s dB_s$, we define the stochastic integrals $\int y dX$ as the integral w.r.t. the local martingale X .

$$\mathbb{X}_{s,t} = \int_s^t x_{s,u} dx_u = \frac{1}{2} x_{s,t}^2 - \frac{1}{2} (\langle X \rangle_t - \langle X \rangle_s).$$

- When $X = B^H$ is a fractional Brownian motion, we define the stochastic integral $\int y \delta B^H$ in the sense of Skorohod-Wick-Itô by using the Wick product. Then we can compute explicitly

$$\mathbb{X}_{s,t} := \int_s^t B_{s,u}^H \delta B_u^H = \frac{1}{2} (B_{s,t}^H)^2 - \frac{1}{2} (t^{2H} - s^{2H}).$$

Universal approximation theorems



Distribution Regression for Sequential Data

Maud Lemerrier¹, Cristopher Salvat², Theodoros Damoulas¹, Edwin V. Bonilla³, and Terry Lyons²

Theorem 2.1. Let $\mathcal{X} \subset \mathcal{C}(I, E)$ be a compact set of paths and consider a continuous function $f : \mathcal{X} \rightarrow \mathbb{R}$. Then for any $\epsilon > 0$ there exists a truncation level $n \geq 0$ such that for any path $x \in \mathcal{X}$

$$\left| f(x) - \sum_{k=0}^n \sum_{J \in \{1, \dots, d\}^k} \alpha_J S(x)^J \right| < \epsilon, \quad (7)$$

where $\alpha_J \in \mathbb{R}$ are scalar coefficients.

Theorem 3.2. Let $\mathcal{X} \subset \mathcal{C}(I, E)$ be a compact set of paths and consider a weakly continuous function $F : \mathcal{P}(\mathcal{X}) \rightarrow \mathbb{R}$. Then for any $\epsilon > 0$ there exists a truncation level $m \geq 0$ such that for any probability measure $\mu \in \mathcal{P}(\mathcal{X})$

$$\left| F(\mu) - \sum_{k=0}^m \sum_{J \in \{1, \dots, d\}^k} \alpha_J S(\Phi(\mu))^J \right| < \epsilon, \quad (14)$$

where $\alpha_J \in \mathbb{R}$ are scalar coefficients.

Signature transform

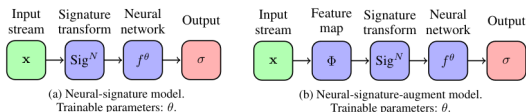


Figure 1: Two simple architectures with a signature layer.

3.6 Inverting the truncated signature

How well does a truncated signature encode the original stream of data? A simple experiment is to attempt to recover the original stream of data given its truncated signature. We remark that finding a mathematical description of this inversion is a challenging task [28, 29, 30].

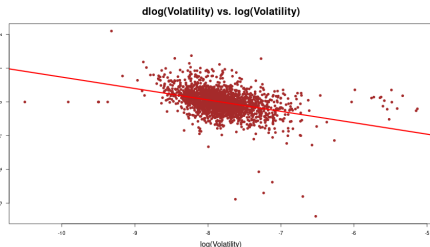
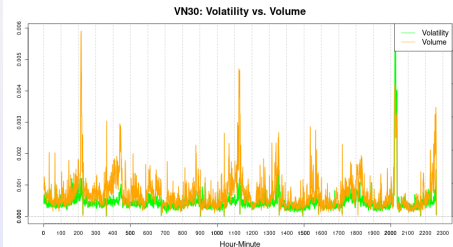
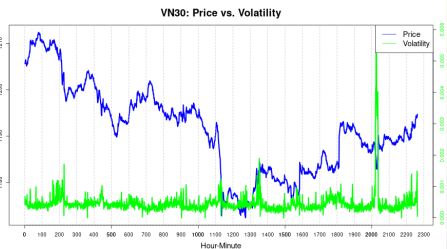
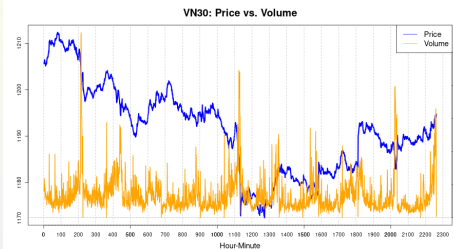
Fix a stream of data $\mathbf{x} = (x_1, \dots, x_n) \in \mathcal{S}(\mathbb{R}^d)$. Assume that the truncated signature $\text{Sig}^N(\mathbf{x})$ and the number of steps $n \in \mathbb{N}$ are known. Now apply gradient descent to minimise

$$L(\mathbf{y}; \mathbf{x}) = \|\text{Sig}^N(\mathbf{y}) - \text{Sig}^N(\mathbf{x})\|_2^2 \quad \text{for } \mathbf{y} = (y_1, \dots, y_n) \in \mathcal{S}(\mathbb{R}^d).$$

Figure 3 shows four handwritten digits from the PenDigits dataset [31]. The solid blue path is the original path $\mathbf{x}$, whilst the dashed orange path is the reconstructed path $\mathbf{y}$ minimising $L(\mathbf{y}; \mathbf{x})$. Truncated signatures of order $N = 12$ were used for this task. We see that the truncated signatures have managed to encode the input paths $\mathbf{x}$ almost perfectly.



Rough volatility model: VN30 realtime data



Stock price models

- ① Samuelson stock model- 1965, Hull-White model - 1987:

$$dS_t = \mu_t S_t dt + \sigma_t S_t dB_t, \quad (3.1)$$

for a stock price S_t , using Itô calculus. σ_t satisfies a stochastic differential equation

$$d \log \sigma_t = k(\theta - \log \sigma_t) dt + \gamma dW_t \quad (3.2)$$

with parameters k, θ, γ and another Brownian motion W_t , with an instantaneous correlation $dB_t dW_t = \rho dt$ with a parameter $\rho \in [0, 1]$ between the two different Brownian motions

- ② Heston model - 1993: σ_t^2 follows a Cox-Ingersoll-Ross equation

$$d(\sigma_t^2) = k \left[\theta - (\sigma_t^2) \right] dt + \lambda \sigma_t dW'_t \quad (3.3)$$

where $k, \theta, \lambda > 0$ are parameters, and the two Brownian motions again possess an instantaneous correlation $dW_t dW'_t = \rho dt$ with $\rho \in [0, 1]$.

Memory effect

- ① Comte-Renault - 1998: ordinary Brownian motion is replaced by fractional Brownian motion in the model (3.2) for the variance σ_t , resulting in

$$d \log \sigma_t = k(\theta - \log \sigma_t)dt + \gamma dB_t^H. \quad (3.4)$$

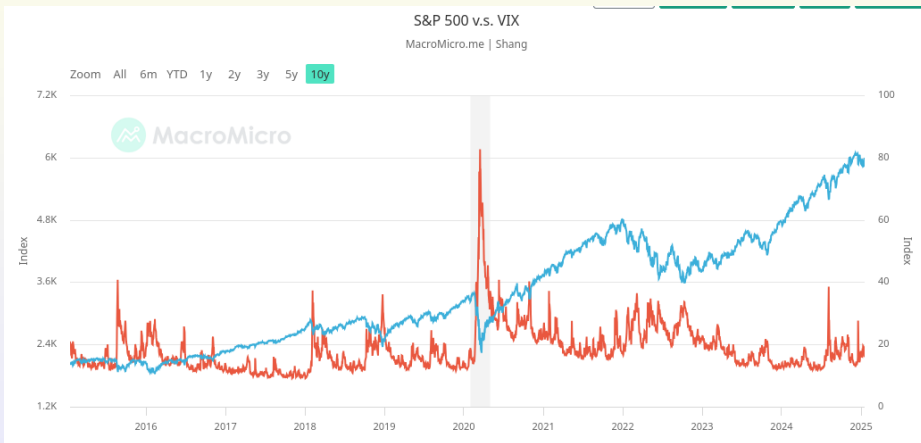
In order to obtain a process with long memory, a Hurst exponent $H > \frac{1}{2}$ is needed.

- ② Gatheral et al. - (2016,2018): if we assume the model (3.4), the log-volatility behaves essentially as a fractional Brownian motion with Hurst exponent H of order 0.1 at any reasonable time scale. Volatility is rough!
- ③ Using Hairer's regularity structure theory, Bayer et al. 2019 considers

$$\begin{aligned} \frac{dS_t}{S_t} &= f(Z_t)(\rho dW_t + \sqrt{1 - \rho^2} dW'_t) \\ Z_t &= z + \int_0^t K(s, t)v(Z_s)ds + \int_0^t K(s, t)u(Z_s)dW_s, \end{aligned} \quad (3.5)$$

- ④ Duc-Jost- 2023: consider first equation, using rough path theory (different rough path lift), not Itô calculus. No arbitrage is proved under transaction cost.

Price vs. implied volatility



Long memory and multi-scaling phenomenon

	1M	5M	15M	1H	4H	Daily	Weekly
Sp500	0.5158	0.5103	0.5234	0.5201	0.5278	0.5613	0.5816
Dow Jones	0.5199	0.5177	0.5301	0.5437	0.5534	0.5764	0.5952
Nasdaq100	0.5177	0.5193	0.5245	0.5324	0.5490	0.5694	0.6069

Table: Hurst exponents for different time scales using [R/S] analysis.

	1M	5M	15M	1H	4H	Daily	Weekly
Sp500	0.4898	0.4911	0.4969	0.4838	0.4818	0.5649	0.6006
Dow Jones	0.4937	0.4952	0.5021	0.4927	0.4951	0.5049	0.5386
Nasdaq100	0.4907	0.4964	0.5048	0.5031	0.4755	0.5188	0.5651

Table: Hurst exponents for different time scales using Spectral analysis.

	1M	5M	15M	1H	4H	Daily	Weekly
Sp500	0.5152	0.5202	0.5157	0.5181	0.5149	0.5549	0.5410
Dow Jones	0.5201	0.5208	0.5174	0.5161	0.5081	0.5651	0.5734
Nasdaq100	0.5271	0.5294	0.5224	0.5307	0.5288	0.6112	0.6490

Table: Hurst exponents for different time scales using Higuchi method.

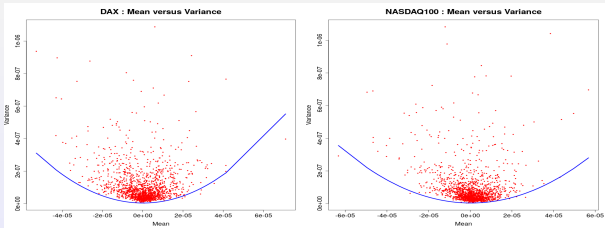
Quadratic relation between variance and expectation of the logarithmic return

A drawback of model (3.1) is the fact that

$$\mathbb{E} Y_{t,t+h} = h\left(\mu - \frac{\sigma^2}{2}\right), \quad \text{Var } Y_{t,t+h} = \sigma^2 h,$$

which implies that the variance depends linearly and negatively on the expected return

$$\text{Var } Y_{t,t+h} = 2\mu h - 2\mathbb{E} Y_{t,t+h}. \quad (3.6)$$



Time dependent rough model

- Accordingly, given parameters $\frac{1}{2} > \alpha > \beta > 0$ which also satisfy $2\alpha + \beta > 1$, let us consider the time dependent model

$$dS_t = \mu_t S_t dt + \sigma_t S_t dx_t \quad (3.7)$$

where $\mu : [0, T] \rightarrow \mathbb{R}$ is Lebesgue integrable and $\sigma \in C^\beta([0, T], \mathbb{R})$ is controlled by a realization $(\omega, x)^T \in C^\beta([0, T], \mathbb{R}) \oplus C^\alpha([0, T], \mathbb{R})$. Equation (3.7) is understood in the integral form

$$S_t = S_0 + \int_0^t \mu_u S_u du + \int_0^t \sigma_u S_u dx_u. \quad (3.8)$$

- The explicit pathwise solution of (3.7) is given by $S_t = e^{Y_t}$ where

$$Y_b = Y_a + \int_a^b \mu_u du - \frac{1}{2} \int_a^b \sigma_u^2 d[x]_u + \int_a^b \sigma_u dx_u \quad (3.9)$$

- Example: When $X = \hat{B}$ is the G -Brownian motion with respect to the sublinear expectation $\mathbb{E}$, then it follows from Peng that $[\hat{B}] = \langle \hat{B} \rangle$ and one can solve the stochastic differential equation

$$S_t = S_0 + \int_0^t \mu S_u du + \int_0^t \sigma S_u d\hat{B}_u \quad (3.10)$$

explicitly as

$$Y_{s,t} = \mu(t-s) + \sigma \hat{B}_{s,t} - \frac{1}{2} \sigma^2 \left(\langle \hat{B} \rangle_t - \langle \hat{B} \rangle_s \right). \quad (3.11)$$

Multi-scaling phenomenon and quadratic relation

$$Y_{t,t+h} = \mu h + \sigma X_{t,t+h} - \frac{\sigma^2}{2} [X]_{t,t+h}, \quad (3.12)$$

where X and $[X]$ are independent. As a result,

$$EY_{t,t+h} = \mu h - \frac{\sigma^2}{2} E[X]_{t,t+h}, \quad \text{Var } Y_{t,t+h} = \sigma^2 \text{Var } X_{t,t+h} + \frac{\sigma^4}{4} \text{Var } [X]_{t,t+h}. \quad (3.13)$$

$$\text{Var } X_{t,t+h} = E|X_{t,t+h}|^2 = Ch, \quad \text{Var } [X]_{t,t+h} = C_H h^{2H} \quad (3.14)$$

It follows from (3.13) that

$$\log(\text{Var } Y_{t,t+h}) = \log\left(C\sigma^2 h + \frac{C_H}{4}\sigma^4 h^{2H}\right) \approx \begin{cases} \log(h) + \log(C\sigma^2) & \text{if } h \ll 1 \\ 2H \log(h) + \log(\frac{C_H}{4}\sigma^4) & \text{if } h \gg 1 \end{cases}. \quad (3.15)$$

$$\text{Var } Y_{t,t+h} = 2 \frac{\text{Var } X_{t,t+h}}{E[X, 2]_{t,t+h}} (\mu h - EY_{t,t+h}) + \frac{\text{Var } [X, 2]_{t,t+h}}{(E[X, 2]_{t,t+h})^2} (\mu h - EY_{t,t+h})^2. \quad (3.16)$$

In particular, since $\sigma^2 = \frac{2}{E[X, 2]_{t,t+h}} (\mu h - EY_{t,t+h}) \geq 0$, it follows that

$$\begin{aligned} \text{Var } Y_{t,t+h} &\geq \frac{\text{Var } [X]_{t,t+h}}{(E[X]_{t,t+h})^2} (\mu h - EY_{t,t+h})^2 \\ \Leftrightarrow (EY_{t,t+h}, \text{Var } Y_{t,t+h}) &\in \mathcal{P} := \left\{ (x, y) \in \mathbb{R}^2 : y = \frac{\text{Var } [X]_{t,t+h}}{(E[X]_{t,t+h})^2} (\mu h - x)^2 \right\}. \end{aligned} \quad (3.17)$$

Rough volatility models

- Stock model:

$$\begin{aligned}dS_t &= S_t(\mu_t dt + \sigma_t dB_t) \\ \sigma_t &= e^{X_t} \\ dX_t &= \alpha(m - X_t)dt + \nu dB_t^H\end{aligned}\tag{3.18}$$

where B is a standard Brownian motion, B^H is a fractional Brownian motion with Hurst index H such that $\text{cov}(B, B^H) = \rho$.

- Data source: realtime data of Vietnam stock index VN30. Order book.
- Discretization scheme

$$\Delta X_t = -\alpha(X_t - m)\Delta t + \nu \Delta W_t^H\tag{3.19}$$

or in the linear form

$$\Delta X_t = c_0 + c_1 X_t + \epsilon_t,\tag{3.20}$$

where

$$c_0 = \alpha m \Delta t, c_1 = -\alpha \Delta t, \epsilon_t = \nu \Delta W_t^H.\tag{3.21}$$

- Hurst index H is estimated using moment method

$$\mathbb{E}[|\Delta W_t^H|^q] \sim (\Delta t)^{qH} \Rightarrow \log \mathbb{E}[|\epsilon_t|^q] = qH \log(\Delta t)\tag{3.22}$$

Learning results: Instability of the model

q	$\zeta(q)$	Hurst
0.5	0.0493	0.0986
1.0	0.0913	0.0913
1.5	0.1160	0.0779
2.0	0.1157	0.0578
2.5	0.0908	0.0395
3.0	0.0818	0.0273

q	$\zeta(q)$	Hurst
0.5	0.0757	0.1473
1.0	0.1238	0.1238
1.5	0.1368	0.0912
2.0	0.1192	0.0598
2.5	0.0970	0.0308
3.0	0.0837	0.0279

q	$\zeta(q)$	Hurst
0.5	0.0300	0.0701
1.0	0.0595	0.0595
1.5	0.0785	0.0527
2.0	0.0218	0.0399
2.5	0.0189	0.0043
3.0	0.0075	0.0018

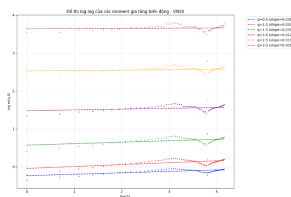
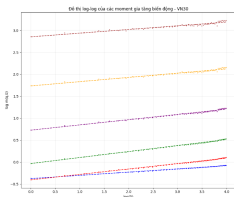
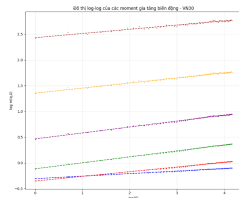
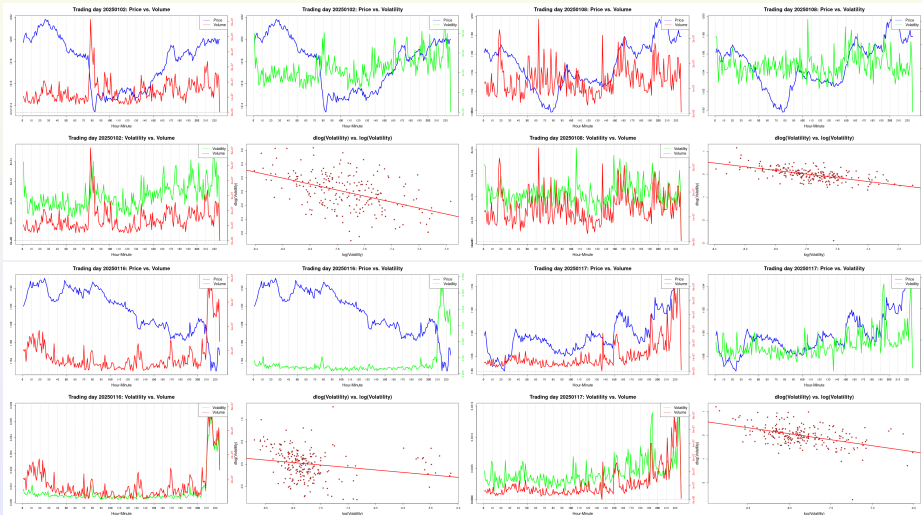


Figure: Different time scale Hurst index

- Instability of the result when time scale is changed.
- When combining data w.r.t. the time scale parameters, it leads to repetition of the data and might create bias.

Empirical evidence

Learning results: model with trading volume



Price signature method

- The problem is to find $\ell^* \in \mathbb{R}^{d^*}$ such that

$$\ell^* \in \arg \min_{\ell} L_{\text{price}, \alpha}(\ell), \quad \text{where}$$

$$\begin{aligned} L_{\text{price}, \alpha}(\ell) &:= \sum_{i=1}^N (S_n(\ell) t_i - S_{t_i})^2 + \alpha(\ell) \\ &= \sum_{i=1}^N \left(S_0 + \ell_0 \langle \mathbf{e}_1, \widehat{\mathbf{X}}_{t_i} \rangle + \sum_{0 < |I| \leq n-1} \ell_I \langle \tilde{\mathbf{e}}_I, \widehat{\mathbf{X}}_{t_i} \rangle - S_{t_i} \right)^2 + \alpha(\ell), \end{aligned}$$

Price signature method

$$\begin{aligned}dS_t &= \mu(S_t, V_t)dt + g(S_t, V_t)dB_t^{\mathbb{P}} \\dV_t &= h(S_t, V_t)dt + \sigma(S_t, V_t)dW_t^{\mathbb{P}}\end{aligned}$$

with $d[B^{\mathbb{P}}, W^{\mathbb{P}}]_t = \rho dt$, where $\rho \in [-1, 1]$ and μ, h, g, σ functions from $\mathbb{R}^2$ to $\mathbb{R}$ such that g and σ are strictly positive. Choosing $d = 2$, we aim to estimate the trajectories of a primary process $X_t := (B_t^{\mathbb{Q}}, W_t^{\mathbb{Q}})$ specified by

$$(4.2) \quad B_t^{\mathbb{Q}} = \int_0^t \frac{\mu(S_s, V_s)}{g(S_s, V_s)} ds + B_t^{\mathbb{P}}, \quad W_t^{\mathbb{Q}} = \int_0^t \frac{h(S_s, V_s)}{\sigma(S_s, V_s)} ds + W_t^{\mathbb{P}}.$$

Note that under sufficient integrability conditions on the market prices of risk, $\mathbb{Q}$ is an equivalent local martingale measure, under which $(B^{\mathbb{Q}}, W^{\mathbb{Q}})$ are correlated $\mathbb{Q}$ -Brownian motions and under which the dynamics of (S, V) are of the following form:

$$\begin{aligned}dS_t &= g(S_t, V_t)dB_t^{\mathbb{Q}}, \\dV_t &= \sigma(S_t, V_t)dW_t^{\mathbb{Q}}.\end{aligned}$$

Let us remark that here it is actually easier to work with the market correlated Brownian motion instead of independent ones, as one can avoid explicitly estimating the correlation

spot covariance estimators in general semimartingale contexts), to get an estimator of $\Sigma_{11} := g(S_t, V_t)^2$ and $\Sigma_{22} := \sigma(S_t, V_t)^2$, which in turn allows computing

$$(4.3) \quad dB_t^{\mathbb{Q}} = \frac{dS_t}{\sqrt{\Sigma_{11,t}}}, \quad dW_t^{\mathbb{Q}} = \frac{dV_t}{\sqrt{\Sigma_{22,t}}}.$$

Observe that assuming g and σ to be strictly positive we get that $\sqrt{\Sigma_{11,t}} = g(S_t, V_t)$ and $\sqrt{\Sigma_{22,t}} = \sigma(S_t, V_t)$. This is important to guarantee that $dS_t = g(S_t, V_t)dB_t^{\mathbb{Q}}$ (as requested in Corollary 3.12) and $dV_t = \sigma(S_t, V_t)dW_t^{\mathbb{Q}}$.



- Discrete systems need not be a discretization scheme of the old form of continuous limiting equations. Probably a new form!

$$\begin{aligned} \varphi^\Delta(0, \omega)y_0 &= y_0, \\ \varphi^\Delta(k\Delta, \omega)y_0 &:= H^\Delta(\theta_{(k-1)\Delta}\omega) \circ \dots \circ H^\Delta(\omega)y_0, \quad \forall k \geq 1. \end{aligned} \quad (3.25)$$

Suggestion: nonlinear model using RNN

- Need to replace the volatility model by a nonlinear model

$$dX_t = f(X_t)dt + \nu dB_t^H \quad (3.26)$$

or in the discretization scheme

$$\Delta X_t = f(X_t)\Delta t + \nu \Delta W_t^H \quad (3.27)$$

- f is learnt using RNN. The residual error ϵ_t is used to learn H .
- Or the model for σ is wrong. σ might also depend on another factor like the trading volume.
- Duc-Jost (SIAM Math. Finance 2023): for $B \perp B^H$

$$\begin{aligned} dS_t &= S_t(\mu_t dt + \sigma_t dB_t) \\ d\sigma_t &= \sigma_t(\rho dB_t + \frac{\sqrt{1-\rho^2} dB_t^H}{1 + e^{\sqrt{1-\rho^2} B_t^H}}). \end{aligned} \quad (3.28)$$

Solving using different rough path lifts (W, B) where $dW_t = \alpha(1 - W_t)dB_t^H$ or

$$W_t = h(B^H) = \frac{e^{\sqrt{1-\rho^2} B_t^H}}{1 + e^{\sqrt{1-\rho^2} B_t^H}}. \text{ Then } \sigma_t = W_t e^{\rho B_t}.$$

Random dynamical systems

A random dynamical system $\varphi : \mathbb{R} \times \Omega \times \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $(t, \omega, y_0) \mapsto \varphi(t, \omega)y_0$ is a measurable mapping w.r.t. (t, ω, y_0) which satisfies the cocycle property

$$\varphi(t+s, \omega)y_0 = \varphi(t, \theta_s \omega) \circ \varphi(s, \omega)y_0, \quad \forall t, s \in \mathbb{R}, \omega \in \Omega, y_0 \in \mathbb{R}^2. \quad (4.1)$$

Generation of an RDS from rough differential equations requires probabilistic setting. Denote by $T_1^2(\mathbb{R}^m) = 1 \oplus \mathbb{R}^m \oplus (\mathbb{R}^m \otimes \mathbb{R}^m)$ the set with the group product

$$(1, g^1, g^2) \bullet (1, h^1, h^2) = (1, g^1 + h^1, g^1 \otimes h^1 + g^2 + h^2),$$

for all $\mathbf{g} = (1, g^1, g^2)$, $\mathbf{h} = (1, h^1, h^2) \in T_1^2(\mathbb{R}^m)$. Then it can be shown that $(T_1^2(\mathbb{R}^m), \bullet)$ is a topological group.

Given $\alpha \in (\frac{1}{3}, \nu)$, assign $\Omega := C_0^{0, \alpha}(\mathbb{R}, T_1^2(\mathbb{R}^m))$ and equip it with the Borel σ -algebra $\mathcal{F}$. Denote by θ the *Wiener-type shift*

$$(\theta_t \omega)_\cdot = \omega_{t+\cdot}^{-1} \bullet \omega_{t+\cdot}, \quad \forall t \in \mathbb{R}, \omega \in C_0^{0, \alpha}(\mathbb{R}, T_1^2(\mathbb{R}^m)), \quad (4.2)$$

and define the so-called *diagonal process* $\mathbf{X} : \mathbb{R} \times \Omega \rightarrow T_1^2(\mathbb{R}^m)$, $\mathbf{X}_t(\omega) = \omega_t$ for all $t \in \mathbb{R}, \omega \in \Omega$. $(\Omega, \mathcal{F}, \mathbb{P})$ is a probability space equipped with the continuous (thus measurable) *metric dynamical system* $\theta : \mathbb{R} \times \Omega \rightarrow \Omega$. In particular, the Wiener shift (4.2) implies that

$$\|\mathbf{x}(\theta_h \omega)\|_{p\text{-var}, [s, t]} = \|\mathbf{x}(\omega)\|_{p\text{-var}, [s+h, t+h]}, \quad N_{[s, t]}(\mathbf{x}(\theta_h \omega)) = N_{[s+h, t+h]}(\mathbf{x}(\omega)). \quad (4.3)$$

Discrete time set $\Pi = \{t_k\}_{k \in \mathcal{N}}$ defined by

$$y_{t_0} \in \mathbb{R}^d, \quad y_{t_{k+1}} = y_{t_k} + f(y_{t_k})(t_{k+1} - t_k) + F_{t_k, t_{k+1}}(y, \mathbf{x}), \quad (4.4)$$

$$F_{s,t}(y, \mathbf{x}) = \begin{cases} g(y_s)x_{s,t}, & \mathbf{x} = x \\ g(y_s)x_{s,t} + Dg(y_s)g(y_s)\mathbb{X}_{s,t}, & \mathbf{x} = (x, \mathbb{X}) \end{cases} \quad (4.5)$$

(H_f) $f \in C(\mathbb{R}^d, \mathbb{R}^d)$ of linear growth, i.e.

$$\exists C_f > 0 : \quad \|f(y)\| \leq C_f \|y\| + \|f(0)\|, \quad \forall y \in \mathbb{R}^d; \quad (4.6)$$

(H_g) either $g \in C_b^2(\mathbb{R}^d, \mathbb{R}^{d \times m})$ such that

$$\|g\|_\infty, C_g := \max \left\{ \|Dg\|_\infty, \|D^2g\|_\infty \right\} < \infty, \quad (4.7)$$

or $g(y) = Cy + g(0)$, $C \in (\mathbb{R}^d, \mathbb{R}^{d \times m})$ s.t. $\|C\| \leq C_g$. For YDE, $g \in C^1$ with bounded derivative.

Theorem (LHD, Kloeden -2023)

Consider the RDE

$$d\hat{y}_t = f(\hat{y}_t)dt + g(\hat{y}_t)dX_t, \hat{y}_a \in \mathbb{R}^d, t \in [a, b] \quad (4.8)$$

where X is a stochastic process with stationary increment which almost all realizations x are of local Hölder continuous with exponent in $(1/3, 1/2]$ such that x is truly rough.

If in addition to $(\mathbf{H}_f)$, $(\mathbf{H}_g)$, f and D^2g are local Lipchitz, then there exists a generic random variable $\hat{\xi}(\hat{y}_a, \mathbf{x}, [a, b])$ such that the following estimate holds

$$\begin{aligned} & \sup_{0 \leq k \leq m} \|\hat{y}(t_k, y_a, x) - y_{t_k}\| \\ & \leq \hat{\xi}(\hat{y}_a, \mathbf{x}, [a, b]) (|\Pi[a, b]| + \|\mathbf{x}\|_{p, [a, b], |\Pi[a, b]|})^{3-p} (b - a + \|\mathbf{x}\|_{p, [a, b]}^p). \end{aligned} \quad (4.9)$$

In particular,

$$\lim_{|\Pi[a, b]| \rightarrow 0} \sup_{0 \leq k \leq m} \|\hat{y}(t_k, \hat{y}_a, x) - y_{t_k}\| = 0. \quad (4.10)$$

Discrete RDS

- Continuous time: Bailleul-Riedel-Scheutzow (2017) proves that RDE $dy_t = f(y_t)dt + g(y_t)dx_t$ generates a continuous RDS $\varphi(t, \omega, y_0)$ on $\mathbb{R}^d$ which is the solution of the RDE w.r.t. the driving path $x(\omega)$ and at initial point y_0 .
- Discrete time: Define the discrete mapping $H^\Delta(\mathbf{x})y = y + f(y)\Delta + F_{0,\Delta}(y, \mathbf{x})$. In LHD-Kloeden (2023), we prove that the discrete Euler scheme (4.4) generates a discrete random dynamical system $\varphi^\Delta : \Pi^\Delta \times \Omega \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ over $(\Omega, \mathcal{F}, \mathbb{P}, \theta)$ such that

$$\begin{aligned}\varphi^\Delta(0, \omega)y_0 &= y_0, \\ \varphi^\Delta(k\Delta, \omega)y_0 &:= H^\Delta(\theta_{(k-1)\Delta}\omega) \circ \dots \circ H^\Delta(\omega)y_0, \quad \forall k \geq 1.\end{aligned}\tag{4.11}$$

- Throughout this paper, we assume that θ is an ergodic dynamical system. In LHD-PT.Hong-ND.Cong (2024), we prove the ergodicity for fBm B^H .

Random pullback attractors

Next, recall that a set $\hat{M} := \{M(\omega)\}_{\omega \in \Omega}$ is called a *random set*, if $\omega \mapsto d(y|M(\omega)) := \inf\{d(y, z) | z \in M(\omega)\}$ is $\mathcal{F}$ -measurable for each $y \in \mathbb{R}^d$. A *universe* $\mathcal{D}$ is a family of random sets which is closed w.r.t. inclusions (i.e. if $\hat{D}_1 \in \mathcal{D}$ and $\hat{D}_2 \subset \hat{D}_1$ then $\hat{D}_2 \in \mathcal{D}$). In our situation, we define the universe $\mathcal{D}$ to be a family of *tempered* random sets $D(\omega)$, that is, $D(\omega)$ is contained in a ball $B(0, \rho(\omega))$ a.s., where the radius $\rho(\omega) > 0$ is a tempered random variable (i.e. $\lim_{t \rightarrow \pm\infty} \frac{1}{t} \log^+ \rho(\theta_t \omega) = 0$ a.s.).

A random set $\mathcal{A}$ is said to be *invariant* if $\varphi(t, \omega)\mathcal{A}(\omega) = \mathcal{A}(\theta_t \omega)$ for all $t \in \mathbb{R}$, $\omega \in \Omega$. An invariant random compact set $\mathcal{A} \in \mathcal{D}$ is called a *pullback attractor* in $\mathcal{D}$, if

$$\lim_{t \rightarrow \infty} d_H(\varphi(t, \theta_{-t} \omega) \hat{D}(\theta_{-t} \omega) | \mathcal{A}(\omega)) = 0, \quad (4.12)$$

where $d_H(\cdot|\cdot)$ is the Hausdorff semi-distance, i.e. $d_H(D|A) := \sup_{d \in D} \inf_{a \in A} \|d - a\|$.

A random set $\mathcal{B}^* \in \mathcal{D}$ is called *pullback absorbing* in the universe $\mathcal{D}$ if $\mathcal{B}^*$ absorbs all closed random sets in $\mathcal{D}$, i.e. for any closed random set $\hat{D} \in \mathcal{D}$, there exists a time $t_0 = t_0(\omega, \hat{D})$ such that

$$\varphi(t, \theta_{-t} \omega) \hat{D}(\theta_{-t} \omega) \subseteq \mathcal{B}^*(\omega), \text{ for all } t \geq t_0. \quad (4.13)$$

Then given the universe $\mathcal{D}$ and a compact pullback absorbing set $\mathcal{B}^* \in \mathcal{D}$, there exists a unique pullback attractor $\mathcal{A}(\omega)$ in $\mathcal{D}$, given by

$$\mathcal{A}(\omega) = \bigcap_{t \geq 0} \overline{\bigcup_{s \geq t} \varphi(s, \theta_{-s} \omega) \mathcal{B}^*(\theta_{-s} \omega)}. \quad (4.14)$$

Existence of random attractors

Lemma

Assign

$$\bar{R}_\lambda(\omega) := \sum_{k=0}^{\infty} e^{-k\delta} H(\|\mathbf{x}(\theta_{-k}\omega)\|_{p\text{-var},[-1,0]}) = \sum_{k=0}^{\infty} e^{-k\delta} H(\|\mathbf{x}(\omega)\|_{p\text{-var},[-1-k,-k]}). \quad (4.15)$$

Then $\bar{R}_\lambda(\cdot)$ are finite a.s. such that $\bar{R}_\lambda \in^1$.

Theorem

There exists a pullback attractor $\mathcal{A}$ for the generated random dynamical system φ of the stochastic system such that $|\mathcal{A}(\cdot)| \in^1$. Moreover,

$$\mathcal{A}(\omega) = \bigcap_{n=1}^{\infty} \varphi(n, \theta_{-n}\omega) \bar{\mathcal{B}}^\varepsilon(\theta_{-n}\omega) = \lim_{n \rightarrow \infty} \varphi(n, \theta_{-n}\omega) \bar{\mathcal{B}}^\varepsilon(\theta_{-n}\omega), \quad \forall \omega \in \Omega. \quad (4.16)$$

where $\bar{\mathcal{B}}^\varepsilon$ is given by

$$\bar{\mathcal{B}}^\varepsilon(\omega) := V^{-1}(K_\lambda + \bar{R}_\lambda(\omega) + \varepsilon), \quad \forall \varepsilon \geq 0. \quad (4.17)$$

Existence of continuous attractors

Now we assume further that f is dissipative: there exist $c, d > 0$ so that for all $y \in \mathbb{R}^d$,

$$\langle y, f(y) \rangle \leq c - d\|y\|^2. \quad (4.18)$$

In addition, assume the p -variation norm of $\mathbf{x}$ is of finite moment for all order $k \in \mathcal{N}$, i.e.

$$\mathbb{E} \|\mathbf{x}(\cdot)\|_{p, \Pi[a, b]}^k < \infty, \quad \forall k \in \mathcal{N}, \quad \forall 0 < a < b, \quad a, b \in \Pi. \quad (4.19)$$

Theorem (LHD-2022)

Assume $(H_f^i), (H_g^b), (H_X)$, then there exists a random pullback attractor $\mathcal{A}$ such that $|\mathcal{A}(\cdot)| \in \mathcal{P}$ for any $\rho \geq 1$. If in addition (H_A) and f is global Lipschitz continuous then the random attractor is upper semi-continuous with respect to the noise intensity in the sense that $\mathcal{A} \rightarrow \mathcal{A}$ a.s. (w.r.t. the Hausdorff semi-distance) as $C_g \rightarrow 0$, both in the almost sure and in $\mathcal{L}^p$ senses. Moreover, if f is strictly dissipative then $\mathcal{A}$ is a singleton provided that C_g is sufficiently small.

Semi-continuity of attractors

Theorem (Cong-LHD-Hong-2022)

Assume $(\mathbf{H}_f)$, $(\mathbf{H}_g)$ and the dissipativity condition (4.37) and condition (4.19). Consider system (4.4) with the regular time set Π^Δ , where $\Delta \in (0, 1)$ satisfied the inequality

$$0 < \Delta < 1 \wedge \frac{d}{2C_f^2} \wedge \frac{1}{2d}. \quad (4.20)$$

Then the generated discrete random dynamical system φ^Δ of (4.4) admits a random pullback attractor ${}^\Delta$.

Theorem

Under assumptions $(\mathbf{H}_f)$, $(\mathbf{H}_g)$ and for all Δ small enough, the attractor $\mathcal{A}^\Delta$ and the absorbing set $\mathcal{B}^\Delta$ of (4.4) is uniformly bounded in Δ a.s.

As a result, for L_g small enough, the numeric attractor $\mathcal{A}^\Delta$ converges to the attractor $\mathcal{A}$ in the Hausdorff semi-distance, i.e. $d_H(\mathcal{A}^\Delta | \mathcal{A}) \rightarrow 0$ as $\Delta \rightarrow 0$, a.s.

RDEs and Doss-Sussmann technique

Solve the controlled differential equation

$$dy_t = f(y_t)dt + g(y_t)dx_t, \quad (4.21)$$

we can also interpret equation (4.21) in the integral form

$$y_t = y_0 + \int_0^t f(y_s)ds + \int_0^t g(y_s)dx_s, \quad \forall t \geq 0, \quad (4.22)$$

where the second integral is a rough integral for the rough path y *controlled* by x in the sense of Gubinelli. The techniques to be used are the Doss-Sussmann technique, i.e. solution $\phi(\mathbf{x}, \phi_a)$ of the *pure* rough differential equation

$$d\phi_u = g(\phi_u)dx_u, \quad u \in [a, b], \phi_a \in \mathbb{R}^d \quad (4.23)$$

is C^1 w.r.t. ϕ_a , and $\frac{\partial \phi}{\partial \phi_a}(\cdot, \mathbf{x}, \phi_a)$ is the solution of the linearized system

$$d\xi_u = Dg(\phi_u(\mathbf{x}, \phi_s))\xi_u dx_u, \quad u \in [a, b], \xi_a = Id, \quad (4.24)$$

where $Id \in \mathbb{R}^{d \times d}$ denotes the identity matrix. The Doss-Sussmann transformation $y_t = \phi_t(\mathbf{x}, z_t)$ map solution y_t of (4.21) on a certain interval $[0, \tau]$ to solution z_t of the associate ordinary differential equation

$$\dot{z}_t = \left[\frac{\partial \phi}{\partial z}(t, \mathbf{x}, z_t) \right]^{-1} f(\phi_t(\mathbf{x}, z_t)), \quad t \in [0, \tau], z_0 = y_0. \quad (4.25)$$

Doss-Sussmann technique

$$\begin{aligned} \|\phi\|_{p\text{-var},[a,b]} &\leq 8C_p C_g \|\mathbf{x}\|_{p\text{-var},[a,b]} ; \\ \left\| \frac{\partial \phi}{\partial \phi_a}(t, \mathbf{x}, \phi_a) - Id \right\|, \left\| \left[\frac{\partial \phi}{\partial \phi_a}(t, \mathbf{x}, \phi_a) \right]^{-1} - Id \right\| &\leq 16C_p C_g \|\mathbf{x}\|_{p\text{-var},[a,b]} . \end{aligned} \quad (4.26)$$

Introduce the notations

$$\eta_t := y_t - z_t, \quad \text{and} \quad \psi_t := \left[\frac{\partial \phi}{\partial z}(t, \mathbf{x}, z_t) \right]^{-1} - Id, \quad \forall t \in [0, \tau],$$

where $\tau > 0$ is chosen such that $16C_p C_g \|\mathbf{x}\|_{p\text{-var},[0,\tau]} \leq \lambda$ for some $\lambda \in (0, 1)$ small enough.

With such τ , it then follows from (4.26) that

$$\|\eta_t\| = \|\phi_t(\mathbf{x}, z_t) - z_t\| \leq \frac{\lambda}{2} \quad \text{and} \quad \|\psi_t\| \leq \lambda, \quad \forall t \in [0, \tau]. \quad (4.27)$$

To estimate $\|z_t\|$, we rewrite (4.25) as

$$\dot{z}_t = (Id + \psi_t)f(z_t + \eta_t). \quad (4.28)$$

Lyapunov functions

(H_V) There exists for a Lyapunov-type function V for the drift f which satisfies

- there exists positive definite functions $\alpha, \beta > 0$ such that

$$\alpha(\|z\|) \leq V(z) \leq \beta(\|z\|), \quad \forall z \in \mathbb{R}^d; \quad (4.29)$$

- there exists a constant $L_V > 0$ such that

$$\|\nabla V(z)\| \leq L_V, \quad \forall z \in \mathbb{R}^d; \quad (4.30)$$

- there exists a sufficiently small constant $\lambda \in (0, 1)$ and constants $C_\lambda, \delta > 0$ such that

$$\sup_{\|\psi\|, \|\eta\| \leq \lambda} \langle \nabla V(z), (I + \psi)f(z + \eta) \rangle \leq C_\lambda - \delta V(z), \quad \forall z \in \mathbb{R}^d. \quad (4.31)$$

Condition (4.31) is stronger than the classical condition on the gradient ∇V along the vector field f : there exist constants $d_1, d_2 > 0$ such that

$$\langle \nabla V(z), f(z) \rangle \leq d_1 - d_2 V(z), \quad \forall z \in \mathbb{R}^d. \quad (4.32)$$

(4.31) is designed in such a way that it does not depend on g but only on the control parameter λ that relates to the intensity C_g .

Stability with Lyapunov functions

Theorem

Under the assumptions (H_V) , (H_g^b) and for any realization x satisfying (H_X) , there exists a solution of (4.21) on any interval $[0, T]$. Moreover, there exist a constant $\delta > 0$ such that any sufficiently small $\lambda \in (0, 1)$ is associated with a constant $K_\lambda > 0$ of which the following estimate holds

$$V(y_t) - K_\lambda \leq e^{-\delta t} [V(y_0) - K_\lambda] + L_V \lambda \int_0^t e^{-\delta s} dN^-\left(\frac{\lambda}{16C_p C_g}, \mathbf{x}, [t-s, t]\right), \quad \forall t \in [0, T]. \quad (4.33)$$

Corollary

Under the assumptions of Theorem 7, there exist constants $\delta > 0$ such that for any $\lambda \in (0, 1)$ small enough, there exist a constant $K_\lambda > 0$ such that the following estimate holds for all

$$V(y_t) - K_\lambda \leq e^{-\delta t} [V(y_0) - K_\lambda] + H(\|\mathbf{x}\|_{p\text{-var}, [0, t]}), \quad \forall t \in [0, T] \quad (4.34)$$

where

$$H(\xi) = L_V (16C_p C_g)^p \lambda^{1-p} \xi^p + 8L_V C_p C_g \xi. \quad (4.35)$$

Applications

- Globally Lipschitz continuous drifts

$$\|f(z_1) - f(z_2)\| \leq C_f \|z_1 - z_2\|, \quad \forall z_1, z_2 \in \mathbb{R}^d. \quad (4.36)$$

Then condition (4.32) follows condition (4.31).

- Consider f to satisfy the global Lipschitz continuity and the dissipativity condition, i.e. there exists $d_1 \geq 0, d_2 > 0$ such that

$$\langle z, f(z) \rangle \leq d_1 - d_2 \|z\|^2, \quad \forall z \in \mathbb{R}^d. \quad (4.37)$$

Then construct $V(z) := \sqrt{1 + \|z\|^2}$.

- The damped pendulum system in 2 dimension $z = (v, w) \in \mathbb{R}^2$ given by

$$\begin{cases} dv_t &= w_t \\ dw_t &= -\sigma^2 \sin v_t - 2\mu w_t \end{cases} \quad (4.38)$$

Then f is globally Lipschitz continuous. $V(z) := \sqrt{1 + \frac{1}{2}w^2 + \sigma^2(1 - \cos v)}$.

- Fitzhugh-Nagumo neuro-system

$$\begin{aligned} dv_t &= v_t - \frac{v_t^3}{3} - w_t + I \\ dw_t &= \varepsilon(v_t - \mu w_t + J) \end{aligned} \quad (4.39)$$

Then construct $V(y) := \left(1 + v^4 + \frac{6}{\varepsilon\mu} w^2\right)^{\frac{1}{4}}$.

Stability at equilibrium

For simplicity of presentation, throughout this section we assume that f is globally Lipschitz continuous with $f(0) = 0$, while $g \in C_b^3$ with $g(0) = 0$. Then

$$\|\phi\|_{p-\text{var},[a,b]} \leq 8C_p C_g \|\mathbf{x}\|_{p-\text{var},[a,b]} \min \{\|\phi_a\|, 1\}. \quad (4.40)$$

Given the Doss-Sussmann transformation $y_t = \phi_t(\mathbf{x}, z_t)$, the difference $\eta_t := y_t - z_t$ on each time interval $16C_p C_g \|\mathbf{x}\|_{p-\text{var},[\tau_i, \tau_{i+1}]} \leq \lambda$ then satisfies $\|\eta_t\| \leq \lambda \|z_t\|$ for all $t \in [\tau_i, \tau_{i+1}]$. Moreover η_t is then a Lipschitz continuous function of z_t . This motivates us to define $L(\lambda)$ to be the convex set of Lipschitz continuous function η of z such that $\|\eta(z)\| \leq \lambda \|z\|$.

Therefore, the assumptions for our targeted Lyapunov function V^* are modified with additional conditions that $V^*(0) = 0$ and $C_\lambda = 0$ in (4.31), i.e. there exists $\lambda \in (0, 1)$ sufficiently small and a constant $\mu > 0$

$$\sup_{\|\psi\| \leq \lambda, \eta \in L(\lambda)} \langle \nabla V^*(z), (I + \psi)f(z + \eta(z)) \rangle \leq -\mu V^*(z), \quad \forall z \in \mathbb{R}^d. \quad (4.41)$$

Lyapunov neural networks

Lyapunov neural network $V_\vartheta : \mathbb{R}^d \rightarrow \mathbb{R}_+$ from an arbitrary neural network $N_\vartheta : \mathbb{R}^d \rightarrow \mathbb{R}$ with the set of its κ -trainable parameters $\vartheta \in \mathbb{R}^\kappa$, given by

$$V_\vartheta(z) := |N_\vartheta(z) - N_\vartheta(0)| + \bar{\alpha} \|z\| \quad (4.42)$$

where $\bar{\alpha} > 0$ is a small user-chosen parameters.

Recall that the training process of Lyapunov neural network V_ϑ in (4.42) aims to find a specific network parameter ϑ such that the condition (4.31) holds for V_ϑ at every $z \in \mathbb{D}$ of a compact domain $\mathbb{D}$. This can be achieved by minimize the risk function

$$l(\vartheta) := \frac{1}{|\mathbb{D}|} \int_{\mathbb{D}} \left[\sup_{\|\psi\| \leq \lambda, \eta \in L(\lambda)} \langle \nabla V_\vartheta(z), (I + \psi)f(z + \eta(z)) \rangle + \bar{\mu} V_\vartheta(z) \right]_+ dz \quad (4.43)$$

for a user chosen parameter γ . In practice, the integral (4.43) does not have analytic form, thus we approximate $l(\vartheta)$ using the empirical expectation

$$\hat{l}(\vartheta) := \frac{1}{M} \sum_{i=1}^M \left[\sup_{\|\psi\| \leq \lambda, \eta \in L(\lambda)} \langle \nabla V_\vartheta(z_i), (I + \psi)f(z_i + \eta(z_i)) \rangle + \bar{\mu} V_\vartheta(z_i) \right]_+ \quad (4.44)$$

where $\{z_i : i = 1, \dots, M\}$ are independent and identically distributed (i.i.d) samples from the uniform distribution $U(\mathbb{D})$ on $\mathbb{D}$.

Main Results

Lemma (Universal approximation)

Given $\|h\|_{W^{1,\infty}(\mathbb{D})} < B$ for some $B > 0$ and for all $\varepsilon \in (0, \frac{1}{2})$, there exists a feedforward neural network N_{ϑ} with the network parameter ϑ and the activation function $\text{RePU}(z) := \max\{0, z\}$ such that $\|N_{\vartheta} - h\|_{W^{1,\infty}(\mathbb{D})} \leq \varepsilon$, where

$$\|h\|_{W^{1,\infty}(\mathbb{D})} := \max\{\text{esssup}_{z \in \mathbb{D}} \|h(z)\|, \max_{1 \leq i \leq d} \text{esssup}_{z \in \mathbb{D}} \|D_i h(z)\|\}$$

Lemma (Lya Nets are dense)

For any $\delta > 0$, the set of Lyapunov-Nets of the form (4.42) with RePU network N_{ϑ} and bounded weights $\vartheta \in [-1, 1]^{\kappa}$ is dense in the function space

$\mathcal{S} := \{\bar{h}(z) + \alpha\|z\| : \bar{h} \in C^1(\mathbb{D}, \mathbb{R}_+), \bar{h}(0) = 0\}$ under $W^{1,\infty}(\mathbb{D}_{\delta})$ where $\mathbb{D}_{\delta} = \mathbb{D} \setminus B(0, \delta)$.

Lemma

For any $\delta, c \in (0, 1)$, there exists an integer $M = M(\delta, c) \in \mathcal{N}$ such that for some M sampling points $z_i \in \mathbb{D}_{\delta}$ where $i = 1, \dots, M$ we obtain $\mathbb{D}_{\delta} \subset \bigcup_{i=1}^M B(z_i, c\|z_i\|)$

Main results

Lemma

For any $\varepsilon \in (0, \frac{\mu\alpha}{4C_f+2\mu})$ and $\delta > 0$, there exists $\vartheta^* \in \Theta$ such that V_{ϑ^*} satisfies (4.29) and

$$\sup_{\|\psi\| \leq \lambda, \eta \in L(\lambda)} \langle \nabla V_{\vartheta^*}(z), (I + \psi)f(z + \eta(z)) \rangle \leq -a_{\mu,\varepsilon} V_{\vartheta^*}(z), \quad \forall z \in \mathbb{D}_\delta, \quad (4.45)$$

where $a_{\mu,\varepsilon} := \mu - \frac{\varepsilon}{\alpha - \varepsilon}(\mu + 4C_f) > 0$.

For bounded weights Θ and the fixed network V_{ϑ} , the function $\langle \nabla V_{\vartheta}(y), (I + \psi)f(y + \eta(y)) \rangle$ is also globally Lipschitz continuous w.r.t. y on $\mathbb{D}$, with a global Lipschitz constant $L(\mathbb{D}, \Theta)$ that is independent of ϑ .

Theorem

For any $\delta \in (0, 1)$ and $a_{\mu,\varepsilon}$ as defined in Lemma 12, choose any $\bar{\gamma} \in (0, a_{\mu,\varepsilon})$ and

$0 < c < \frac{\bar{\gamma}(\alpha - \varepsilon)}{L(\mathbb{D}, \Theta)} < 1$ (if $\frac{\bar{\gamma}(\alpha - \varepsilon)}{L(\mathbb{D}, \Theta)} \geq 1$ then choose any $c \in (0, 1)$). Let M and

$Z_M := \{z_i : i = 1, \dots, M\}$ be given as in Lemma 11 and the empirical risk function $\hat{l}(\vartheta)$ as in (4.44).

Then the minimizer of $\hat{l}$ exists and achieves minimal value 0. Moreover, for any minimizer $\hat{\vartheta}$ of $\hat{l}(\cdot)$, the corresponding $V_{\hat{\vartheta}}$ is an δ -accurate Lyapunov function of f on $\mathbb{D}$.

Thank you!